

Project Design Phase

Problem – Solution Fit Template

Date	29 September 2026
Team ID	SWTID-2026-9139
Project Name	AI WeatherWise
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem–Solution Fit identifies the customer problem and checks whether the proposed solution addresses the customer's actual needs. For AI WeatherWise, the target user is a traveller or outdoor-activity user who needs convenient and actionable weather information.

Purpose:

- Understand the existing weather-planning situation of the target user.
- Identify the user's problems, constraints, behaviour and triggers.
- Connect the identified problems with a practical backend solution.
- Determine how the solution can fit the user's needs and available channels.

Template:

The completed Problem–Solution Fit canvas below follows the same section structure shown in the provided template and applies it to AI WeatherWise.

Problem – Solution Fit Canvas

AI WeatherWise

1. CUSTOMER SEGMENT(S) Who is your customer?	6. CUSTOMER CONSTRAINTS What constraints prevent customers from taking action?	5. AVAILABLE SOLUTIONS Which solutions are available to customers when they face the problem?
Primary users: • Travellers planning trips • People planning outdoor activities • Users checking weather for multiple cities Need: convenient, understandable and actionable weather information.	• Time required to check several locations separately • Raw weather data may require interpretation • Changing weather conditions affect plans • Dependence on external weather/AI services • Users need secure handling of personal preferences and saved locations	• General weather websites/apps • Mobile weather applications • Search engines and weather widgets • Manual comparison of weather information from multiple locations Gap: raw data may not provide personalized activity/clothing guidance.
2. JOBS-TO-BE-DONE / PROBLEMS	9. PROBLEM ROOT CAUSE	7. BEHAVIOUR
• Check current weather before travelling or going outdoors • Compare weather across locations • Decide what activities are suitable • Decide suitable clothing • Keep frequently used locations available	The root problem is the gap between raw weather information and practical decision-making. Users may need to gather information from different places and interpret weather conditions themselves before deciding what to do.	• Searches weather before activities or travel • Checks multiple locations when needed • Looks for temperature, humidity, wind and conditions • Wants recommendations that can be acted upon • May save favorite locations for repeated access
3. TRIGGERS	10. YOUR SOLUTION	8. CHANNELS OF BEHAVIOUR
• Planning a trip • Preparing for an outdoor activity • Sudden change in weather • Need to check another city • Need for clothing/activity guidance	AI WeatherWise: a modular RESTful backend using Node.js and Express.js with MongoDB/Mongoose. It provides authenticated access to current weather information, favorite-location management and weather-based recommendations, with fallback support when external services are unavailable.	Online: REST API requests through a client such as Postman/Thunder Client; weather information and recommendations are retrieved through API endpoints. Offline/physical behaviour: the user's travel or outdoor planning decisions occur outside the system and are informed by the returned weather information.
4. EMOTIONS: BEFORE / AFTER		
Before: uncertain, inconvenient, time-consuming when checking several locations. After: more informed, convenient and confident when current weather and practical recommendations are available through one backend service.		

Project Design Phase – Problem–Solution Fit completed for AI WeatherWise.